

Name: _____

Date: _____

Standards: MA.5.AR.1.1; MA.5.AR.3.1-3.2

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 13: ratio.

1 Recall: Definition of Ratio

A ratio compares two quantities.

Write the ratio of 12 circles to 8 squares in three forms and simplify.

12 to 8; 12:8; 12/8. Simplified: 3:2. _____

2 Recall: Equivalent RatiosComplete the equivalent ratio: $4:7 = 16:?$

Show how you scaled up the ratio.

Scale factor = $16 \div 4 = 4$. So $7 \times 4 = 28$. Answer: 28. _____**3 Application: Find a Quantity**

The ratio of wins to losses for a team is 5:2. The team won 35 games.

How many games did they lose? How many total games did they play?

Scale factor = 7. Losses = $2 \times 7 = 14$. Total = $35 + 14 = 49$ games. _____

4 Application: Three Quantities

Flour, butter, and sugar are in the ratio 4:1:2. You have 8 cups of flour.

How much butter and sugar do you need?

1 unit = $8 \div 4 = 2$. Butter = 2 cups; Sugar = 4 cups. _____

5 Word Problem: Sharing

Three cousins share \$280 in the ratio 3:4:7.

How much does each cousin receive?

Total parts = 14. 1 part = \$20. Cousin 1 = \$60; Cousin 2 = \$80; Cousin 3 = \$140. _____

6 Word Problem: Recipe Scale

A granola recipe uses oats to nuts to dried fruit in the ratio 6:2:1. You want to use 18 cups of oats.

How many cups of nuts and dried fruit? What is the total amount?

1 unit = 3. Nuts = 6 cups; Dried fruit = 3 cups. Total = 27 cups. _____

7 Word Problem: Classroom

Students preferring math to reading are in the ratio 5:3. There are 40 students total.

How many prefer math? How many prefer reading?

Total parts = 8. 1 part = 5. Math = 25; Reading = 15. _____

8 Find the Mistake

A student says 9:15 in simplest form is 4:5.

Find the error and write the correct simplified ratio.

$\text{GCF}(9, 15) = 3$. $9 \div 3 : 15 \div 3 = 3:5$. The student used the wrong GCF. Correct: 3:5. _____

9 Multi-Step: Ratio Changes

A fish tank has 8 goldfish and 12 tetras. Goldfish are added until the ratio of goldfish to tetras is 1:1.

How many goldfish need to be added?

For 1:1 with 12 tetras, need 12 goldfish. Add $12 - 8 = 4$ goldfish. _____

10 Reasoning: Can They Both Be Right?

Mira says the ratio of small to large stickers is 3:5. Leo says for every 5 large stickers there are 3 small stickers.

Are they saying the same thing? Explain.

Yes. Both describe the same ratio from different perspectives. They are equivalent descriptions.